

Stackly Brew — Cafe & Coffee Shop Platform

Introduction & System Overview

This project is a premium cafe and coffee shop platform built to deliver a sophisticated, warm digital experience that empowers customers to explore the menu, reserve tables, track orders, and connect with the cafe community. The system seamlessly connects a polished, multi-page public-facing website with powerful internal dashboards, enabling customers to browse coffee offerings, manage reservations, earn rewards, and engage with barista insights — all while maintaining an elegant, artisanal brand aesthetic.

Behind the scenes, the architecture integrates a clean, semantic HTML5 layout with advanced CSS styling — including responsive grid systems, CSS custom properties, and modern layout techniques — alongside robust JavaScript functionality such as dynamic form validation, localStorage session management, interactive accordions, appointment booking workflows, and real-time order tracking. Smooth motion effects powered by GSAP and ScrollTrigger enhance the browsing experience with hardware-accelerated animations, scroll-triggered reveals, and staggered transitions that keep interactions feeling natural, responsive, and visually inviting across every screen size — from mobile phones to high-resolution desktops.

Technology Stack & Dependencies

Core Technologies

HTML5 & CSS3: The foundational building blocks used to structure the semantic page content and style the modern, responsive layout. Utilizes CSS custom properties (variables) to maintain a consistent design system featuring the signature Cream (#FFFBF5), Lavender (#C3ACD0), Purple (#7743DB), and Ink (#241B2F) palette, alongside advanced techniques like CSS Grid for complex editorial layouts and Flexbox for responsive content organization. The design uses

warm, inviting backgrounds with subtle shadows and refined borders to maintain a premium artisanal cafe aesthetic.

Vanilla JavaScript: Lightweight, native scripting utilized to handle real-time user inputs, complex form validation (reservation booking, login, signup, contact forms with error handling), click events, dynamic view switching in dashboards, custom dropdown toggles, interactive accordions, localStorage session management for user authentication, and responsive mobile menu functionality without the need for heavy third-party frameworks.

GSAP (GreenSock Animation Platform): A high-performance animation engine used to power hardware-accelerated visual reveals, smooth transitions, and engaging scroll-triggered effects. Combined with the ScrollTrigger plugin, it delivers staggered entrance animations, floating card effects, continuous orbit rotations, and sophisticated motion sequences that bring the interface to life without compromising performance.

External Libraries & CDNs

GSAP & ScrollTrigger: Loaded externally via CDN to power complex, scroll-driven motion sequences, including staggered card reveals, viewport-triggered animations, floating badge effects, orbit rotations, and immersive transitions that enhance the user experience across all public-facing pages — from the hero section to coffee guides and journal content.

Font Awesome Icons: Included via CDN to provide a vast library of scalable vector icons used consistently across navigation, buttons, feature lists, social media links, and interactive elements throughout the platform.

Google Fonts: Imported to provide clean, modern typography with DM Sans for body copy and DM Mono for technical elements, while system fonts (Georgia, Arial) provide the editorial serif and clean sans-serif hierarchy. Multiple weights ensure optimal readability and typographic flexibility across all screen sizes.

File Architecture & Directory Structure

The project is structured as a Multi-Page Application (MPA), where each separate HTML file serves as an independent, dedicated screen or page view. All pages are

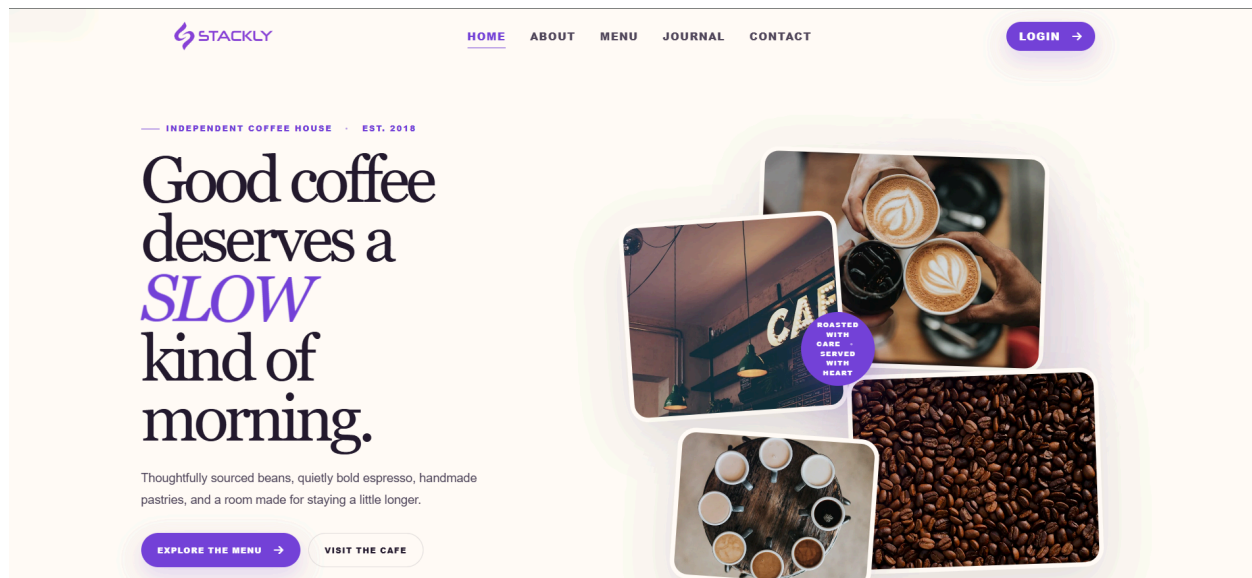
organized within a clean, logical directory structure that separates assets, styles, scripts, and HTML files for maintainability and scalability.

```
stackly-brew/
├─ index.html          # Root entry point
├─ 404.html            # Custom error page
├─ global.css          # Global design tokens & variables
├─ index.css           # Home page styles
├─ logo.webp           # Brand logo asset
├─ structure.txt       # Directory reference
├─ css/               # Page-specific stylesheets
│   └─ about.css       # About page styles
│   └─ blog.css        # Journal page styles
│   └─ contact.css     # Contact page styles
│   └─ login.css       # Login page styles
│   └─ services.css    # Menu page styles
│   └─ signup.css      # Signup page styles
└─ html/              # All HTML pages (excluding root index)
    └─ about.html      # About page
    └─ admin.html      # Admin console
    └─ blog.html       # Journal page
    └─ contact.html    # Contact page
    └─ login.html      # Login page
    └─ services.html   # Menu & Services page
    └─ signup.html     # Signup page
    └─ user.html       # User dashboard
```

Directory Breakdown

Folder	Purpose	Key Files
Root	Core entry points and global styles	index.html, global.css, index.css, logo.webp, 404.html, structure.txt
/css/	Page-specific stylesheets	about.css, blog.css, contact.css, login.css, services.css, signup.css
/html/	All HTML pages (excluding root index)	about.html, admin.html, blog.html, contact.html, login.html, services.html, signup.html, user.html

Visual Design System (UI/UX)



Modern CSS Grid & Flexbox

Implements responsive layouts and complex grid structures via `display: grid` and `display: flex` to seamlessly organize content across all screen sizes. The platform uses custom breakpoints at 1100px, 900px, 850px, 700px, and 480px for optimal device adaptation. Complex coffee menu showcases, masonry journal layouts, reservation forms, and dashboard analytics panels all leverage these modern layout techniques to maintain visual harmony across devices.

Interactive Micro-Animations

Built with smooth CSS transitions (`transition: all 0.4s cubic-bezier(0.22, 1, 0.36, 1)`) and scale transforms (`transform: translateY(-3px)` , `transform: scale(1.02)`) on hover to give immediate visual feedback on buttons, menu cards, coffee items, reservation cards, and clickable elements. These subtle interactions create a responsive, polished user experience that feels natural and intuitive. The floating badge animations and orbit rotations add dynamic motion without overwhelming the interface.

Responsive Mobile-First Adaptations

Implements a full-screen overlay hamburger menu on mobile, collapses complex grids into single-column layouts, adjusts typography sizes and spacing, and transforms the dashboard sidebar into a slide-out menu on smaller screens to ensure optimal readability and usability across all devices. Menu cards, journal

articles, reservation booking forms, and dashboard analytics all adapt gracefully to mobile viewports, ensuring customers can access critical cafe information on the go.

Warm Artisanal Aesthetic

The visual design prioritizes warmth, craftsmanship, and community through:

- **Refined color palette:** Cream (#FFFBF5) for backgrounds, Lavender (#C3ACD0) for accents, Purple (#7743DB) for CTAs, and Ink (#241B2F) for depth
- **Editorial typography:** Georgia serif for display headings conveying craftsmanship, Arial for clean body text
- **Layered compositions:** Asymmetrical grids, overlapping cards, and scroll-driven storytelling
- **Subtle textures:** Noise overlays, soft gradients, and grain patterns for depth
- **Generous whitespace:** Ample padding and margin to create a premium, unhurried experience

Visual Design Language & Color Palette

Color Name	CSS Variable	Hex / RGBA Code	Usage Context
Cream	--cream	#FFFBF5	Main light backgrounds, page backgrounds, cards
Cream Deep	--cream-deep	#F7EFE5	Section backgrounds, alternate surfaces
Lavender	--lavender	#C3ACD0	Accent color, decorative elements, highlights
Purple	--purple	#7743DB	Primary brand color, CTAs, buttons, active states
Ink	--ink	#241B2F	Primary text on light backgrounds, dark sections
Ink Soft	--ink-soft	#51475D	Secondary text, body copy
Muted	--muted	#776D80	Tertiary text, meta information, captions

Color Name	CSS Variable	Hex / RGBA Code	Usage Context
White	--white	#FFFFFF	Card backgrounds, form inputs, content surfaces
Border	--border	rgba(36, 27, 47, 0.12)	Card borders, section dividers
Border Strong	--border-strong	rgba(119, 67, 219, 0.28)	Hover states, emphasized borders
Purple Soft	--purple-soft	rgba(119, 67, 219, 0.08)	Soft backgrounds, badge fills
Lavender Soft	--lavender-soft	rgba(195, 172, 208, 0.28)	Decorative backgrounds, orb fills

Typography System

Font Name	CSS Variable	Font Family	Usage Context
Primary Font	--font-body	Arial, Helvetica, sans-serif	Clean sans-serif for body text, headings, paragraphs, and form inputs
Display Font	--font-display	Georgia, "Times New Roman", serif	Display headings, hero titles, editorial statements
Font Weights	—	400, 500, 600, 700, 800	Multiple weights for clear visual hierarchy from body text to bold headings

Overall Layout & Wireframe

Public-Facing Architecture

Home Page (index.html)

Hero Section: Dynamic hero with four-image reveal, featuring:

- **Left Content:** Badge pill ("Independent Coffee House · Est. 2018"), hero title ("Good coffee deserves a SLOW kind of morning.") with animated wave word, descriptive paragraph, CTA buttons ("Explore the Menu", "Visit the Cafe"), and freshness indicator

- **Right Visual:** Four overlapping photo frames with playful rotations, floating "Roasted with care · served with heart" stamp, and GSAP-powered entrance animations
- **Hero Animations:** Wave word letter animation, staggered photo reveals from different directions, stamp pop-in with back.out easing

Highlights Marquee: Full-width scrolling marquee featuring key specialties (Specialty Coffee, Fresh Pastries, Slow Mornings, Handcrafted Drinks, Local Community) with seamless infinite scroll.

Story Preview Section: Sticky visual storytelling with pinned image and scrolling content blocks:

- Three narrative blocks: "Where it started", "How we roast", "Why it matters"
- Pinned image with caption overlay
- Scroll-triggered content progression

Menu Section: Asymmetrical bento grid featuring four menu cards:

- Espresso (large card with image)
- Milk Coffee (tall card)
- Cold Brew (standard card)
- Pastries & Breakfast (wide card)
- Hover effects with image zoom and overlay gradients

Featured Section: Split layout with overlapping purple card:

- Large featured coffee image
- Purple card with "This Week's Pour" content
- Featured details badges (Seasonal, Single Origin, Barista Pick)

Specialties Section: Four specialty cards with hover effects:

- Stackly Espresso, Butter Croissant, Today's Pour Over, Lavender Cold Brew
- Each with numbered label, title, and image reveal

Why Us Section: Sticky scrolling section with four reasons:

- "We look for character, not just scores"
- "We roast to reveal, never to hide"
- "Precision without the performance"
- "The final ingredient is hospitality"
- Progressive activation with scroll

Team Section: Three team member cards with 3D tilt hover effects:

- Aria (Head Barista), Noah (Roaster), Mira (Cafe Host)
- Image zoom and overlay content

Journey Section: Four-step visit narrative:

- Walk through the door, Choose your cup, Watch it come together, Stay for another
- Timeline with progress dots

Statistics Section: Four animated counters:

- 48,000+ Cups served
- 18 Coffee origins
- 6 Years brewing
- 4.9 Guest rating

Testimonials Section: Infinite scrolling marquee with four testimonial cards:

- Star ratings, quotes, customer names
- Continuous horizontal scroll animation

Journal Section: Editorial article grid:

- One large featured article
- Two supporting articles in side column
- Category badges, read times, and titles

Visit Section: Dark split layout with:

- Cafe interior image

- Address, hours, phone information
- "Get Directions" CTA

Final CTA: Full-width call-to-action with parallax image and "Make room for a good cup" messaging.

Footer: Comprehensive footer with logo, social icons, quick links, hours, and contact details.

Styles: index.css & global.css

Scripts: Embedded JavaScript within index.html

Assets: External Unsplash images, /logo.webp

About Page (/html/about.html)

Hero Section: Full-screen editorial hero with background image, overlay, and large typography ("A cafe built slowly, on purpose."). Features meta information (Est. 2018, The Corner District) and scroll indicator.

Letter Section: Editorial manifesto with:

- Sticky aside with kicker, date, and signature
- Large opening title with drop cap
- Two-column body text
- Rotated stamp decoration

Mission & Vision Section: Diagonal overlapping panels:

- Mission panel (cream) with title, description, and list
- Vision panel (dark) with contrasting design
- Rotated transforms that straighten on hover

Values Section: Expandable accordion with four values:

- Craft, Hospitality, Curiosity, Community
- Click to expand with supporting text
- Animated +/- icon rotation

Journey Section: Horizontal scrolling timeline with five chapters:

- 2018 The First Client
- 2019 The First Roast
- 2021 The Corner District
- 2023 The Second Wave
- 2026 Still Brewing
- Scroll-snap navigation with chapter cards

Team Section: Coverflow carousel with five team members:

- Aria (Head Barista), Noah (Roaster), Mira (Cafe Host), Elias (Pastry Lead), Sana (Green Buyer)
- 3D perspective transforms with active card scaling
- Auto-play with pause on hover
- Dot navigation and prev/next buttons

Philosophy Section: Pull quote editorial with:

- Large opening quote mark
- Featured quote text
- Two-column meta content with principles

Origins Section: Sourcing list with five origins:

- Yirgacheffe (Ethiopia), Huila (Colombia), Nyeri (Kenya), Antigua (Guatemala), Kayanza (Burundi)
- Hover effects with padding shift

Room Section: Tabbed gallery with four spaces:

- The Bar, The Window, The Back Room, The Courtyard
- Tab switching with image and description
- Smooth fade transitions

Final CTA: Overlapping card design with:

- Large image on left

- Card overlapping with "Come be part of the story" content
- Two CTA buttons

Styles: /css/about.css

Scripts: Embedded JavaScript within /html/about.html

Assets: External Unsplash images

Services Page (/html/services.html)

Hero Section: Full-screen hero with clip-path image reveal:

- Large typography ("Everything on the bar.")
- Featured strip with four vertical images
- Marquee ticker at bottom with category names
- GSAP clip-path and scale animations

Overview Section: Bento grid featuring six categories:

- Espresso (tall card), Filter, Cold Brew, Tea, Pastries & Breakfast (wide card), Seasonal
- Hover effects with image zoom and overlay

Signature Section: Sticky showcase with image stack:

- Two overlapping images with parallax
- Rotating "House Signature" stamp
- Product specs (Origin, Process, Notes, Dose)
- Price display

Categories Section: Interactive tabs with five categories:

- Espresso, Filter, Cold Brew, Tea, Food
- Menu lists with prices
- Sticky visual that updates per category

Process Section: Five-step brewing flow:

- Weigh, Grind, Extract, Texture, Serve

- Each step with image and description
- Staggered entrance animations

Food Section: Expandable panels with four categories:

- Pastry, Breakfast, Sweet, Retail Beans
- Each with image, description, and item list
- Click to expand with smooth animation

Seasonal Section: Staggered image cards with four specials:

- Lavender Cold Brew, Espresso Tonic, Maple Oat Latte, Iced Matcha Latte
- Alternating margin offsets for visual interest
- Badge labels (New, Barista Pick)

Catering Section: Split layout with:

- Large catering image with "Booking Now" badge
- Content with three service options
- Enquire CTA

FAQ Section: Accordion with five questions:

- Do you offer decaf?, Are the pastries made in-house?, Can I work from the cafe?, Do you sell beans to take home?, Do you take reservations?
- Sticky aside with media

Final CTA: Full-width parallax with "Come taste the whole menu" messaging.

Styles: /css/services.css

Scripts: Embedded JavaScript within /html/services.html

Assets: External Unsplash images

Journal Page (/html/blog.html)

Hero Section: Editorial masthead with:

- Issue and date information
- Large typography ("Notes from the bar, slowly.")

- Supporting description
- Three thumbnail strip with staggered positioning

Featured Article Section: Large content spotlight:

- Featured image with "Featured" badge
- Category, read time, title, excerpt
- Author byline with avatar
- Read link

Categories Section: Horizontal scroll pills with six categories:

- Coffee Guides (24), Brewing Tips (18), Origin Stories (12), Barista Notes (31), Cafe Life (9), Recipes (15)
- Number, label, and count per category

Latest Articles Section: Masonry grid with eight articles:

- Mixed card sizes (tall, wide, standard)
- Image overlays with category badges
- Title, excerpt, and meta information
- Load More button

Popular Topics Section: Stacked cards with three topics:

- The Latte Reconsidered (#1, 12,400 reads)
- Single Origin, Explained (#2, 9,800 reads)
- Cold Brew vs. Iced Coffee (#3, 8,200 reads)
- Rank, title, description, and read count

Coffee Guides Section: Split layout with:

- Large image with "Read Slowly" stamp
- Content heading and description
- Five-item guide list with hover effects

Barista Insights Section: Three editorial cards:

- Aria on why we stopped adding syrups
- Noah on roast dates and the myth of "fresh"
- Mira on hospitality, and the art of remembering names
- Each with photo, role, title, quote, and read link
- One dark variant for visual contrast

Brewing Resources Section: Interactive tabs with four categories:

- Recipes, Tools, Beans, Water
- Each with intro and list of items

Newsletter Section: Floating card with:

- Large image
- "One good read, every Tuesday" content
- Email form with validation
- Redirect to 404 on submission

Final CTA: Full-width parallax with "Good writing, good coffee" messaging.

Styles: /css/blog.css

Scripts: Embedded JavaScript within /html/blog.html

Assets: External Unsplash images

Contact Page (/html/contact.html)

Hero Section: Split hero with:

- Left content: Badge ("Contact · Visit · Reserve"), title ("Come find us. We are open."), description, CTAs, and open status
- Right visual: Two overlapping photos with "Say Hello" stamp
- GSAP entrance animations

Contact Details Section: Four floating cards:

- Phone, Email, Address, Hours
- Icon, label, value, and supporting note

- Hover lift effects

Reservation Section: Multi-step progressive form:

- Step 1: Who is coming? (Name, Email, Phone)
- Step 2: When are you visiting? (Date, Time, Guests)
- Step 3: Anything else? (Occasion, Notes)
- Terms checkbox and submit
- Progress bar and field validation
- Sticky aside with perks and media

General Inquiry Section: Split layout with:

- Content with heading, description, response time, and media
- Form with Name, Email, Subject, Message, and Terms
- Field validation with error messages

Location Section: Split layout with:

- Google Maps iframe
- Info blocks (Address, Getting Here, Neighbourhood)
- Open in Google Maps CTA

Opening Hours Section: Expandable schedule with:

- Sticky aside with open/closed status
- Five expandable rows (Mon-Thu, Friday, Saturday, Sunday, Public Holidays)
- Each with time and description
- Chevron rotation animation

Private Events Section: Split layout with:

- Image with "Book the Room" badge
- Content with three event options (Full Room Buyout, Back Room Only, Cupping Table)
- Enquire CTA

Departments Section: Interactive tabs with four departments:

- General, Events, Press, Wholesale
- Each with heading, description, and meta list

FAQ Section: Accordion with six questions:

- Do you take walk-ins?, Is there parking nearby?, Are you dog-friendly?, Do you cater outside the cafe?, Can I work from the cafe?, Do you offer gift cards?
- Custom accordion with smooth animation

Final CTA: Full-width parallax with "The kettle is already on" messaging.

Styles: /css/contact.css

Scripts: Embedded JavaScript within /html/contact.html

Assets: External Unsplash images, Google Maps embed

Authentication Portals (/html/login.html & /html/signup.html)

Clean UI: Features clean, visually rich layouts with a split-screen design — left side featuring visual elements (background image, brand content with taglines), right side featuring the authentication form container.

Dynamic Form Controls: Includes interactive input fields with focus highlight effects, custom role selection dropdowns (User/Admin) built with div and button elements, password visibility toggles, and responsive form layouts that adapt to all screen sizes.

Client-Side Validation: Validates required inputs and email formats on submission, triggering error messages for empty fields, invalid email formats, password length requirements (minimum 6 characters), and password confirmation matching on signup.

Role-Based Routing: Evaluates selected account roles (User/Admin) post-validation, dynamically directing users to /html/admin.html or /html/user.html dashboards. The selected role is stored via hidden input and processed on form submission.

Local Storage Session Management: Captures validated credentials upon login, storing the user email into localStorage to retain user identity across pages. The pageshow event ensures user data persists during navigation.

Social Login Placeholders: Provides Google and Apple social login buttons (redirecting to 404.html) for future extensibility.

Navigation Fallbacks: Includes a "Back to Home" link with icon on both login and signup pages, ensuring users can easily navigate back to the main site.

Styles: /css/login.css & /css/signup.css

Scripts: Embedded JavaScript within /html/login.html & /html/signup.html

Assets: /logo.webp

Error Page (404.html)

Brand-Consistent Error Page: Features a clean, minimalist error page with a large "404" display, friendly error title ("Looks like this cup went missing."), and descriptive text ("The page you were looking for has either been moved, renamed, or quietly retired."), maintaining visual consistency with the brand's warm cafe design language.

Clear Navigation Fallback: Includes prominent call-to-action buttons ("Take Me Home" and "Go Back") styled with the brand's primary design tokens to quickly guide lost visitors back to active sections.

Warm Cafe Aesthetic: Utilizes Purple (#7743DB) accents with Cream (#FFFBF5) backgrounds, creating a warm, inviting error experience that aligns with the brand's artisanal positioning.

GSAP Animations: Features staggered entrance animations for header, badge, 404 number, title, description, and actions. Includes floating motion for the illustration card and its inner elements.

Right-Side Illustration Card: Features a purple gradient card with:

- "Still Warm" kicker
- "Meanwhile, the grinder is still running" title
- Coffee cup image
- Four quick navigation links (Homepage, Menu, Journal, Contact)

Styles: Inline styles within 404.html

Scripts: Embedded JavaScript within 404.html

Assets: /logo.webp, External Unsplash image

Internal Architecture (Dashboards)

The dashboards operate on a Single Page Application (SPA) architecture without page reloads. Using dynamic JavaScript event listeners, the implementation toggles element visibility on the fly, updating the view-section display properties to deliver a fast, seamless tab-switching experience.

Admin Dashboard (/html/admin.html)

Top Bar Navigation: Displays the brand logo, chip ("Admin"), interactive menu links (Command Center, Order Flow, Stock Ledger, Roster, Statements), live status indicator, notification bell, user chip, logout button, and mobile menu toggle.

Full-Screen Mobile Drawer: Features a full-screen overlay drawer with:

- Brand header and close button
- User greeting
- Five navigation links with numbers and arrows
- User info and logout button
- Body scroll locking
- Escape key and backdrop click to close

Dynamic Tab Views: Organized into five separate content panels for instant switching:

Command Center View:

- Editorial hero with greeting, note, and action buttons
- Four live stats (Revenue Today, Orders Served, Avg Ticket, Tables Live)
- Live ticker with scrolling feed of orders, alerts, and updates
- Floor plan with 32 clickable seats (color-coded by status)
- Legend chips for seat types
- Alerts widget with four alerts (low stock, equipment, deliveries, bookings)
- On Shift Now widget with four staff members

Order Flow View:

- Four flow stats (In Queue, Avg Wait, On Bar, Served Today)
- Single vertical stream with four status groups:
 - Received (5 orders)
 - Brewing (4 orders)
 - Ready (2 orders)
 - Recently Served (28 orders)
- Each order row with ID, items, table, and time
- Hover effects with padding shift

Stock Ledger View:

- Three stock summary stats (Total SKUs, Low Stock, Stock Value)
- Two-column layout with:
 - Stock stream with seven entries (deliveries, usage, spoilage)
 - Sticky sidebar with current balance for seven items
 - Progress bars with warn/danger states
 - Editorial note about stock status

Roster View:

- Seven-day weekly grid with shift cards
- Today highlighted with purple border
- Each shift card with time, name, avatar, and role
- Leaderboard with five team members
- Rating, drinks count, and rank

Statements View:

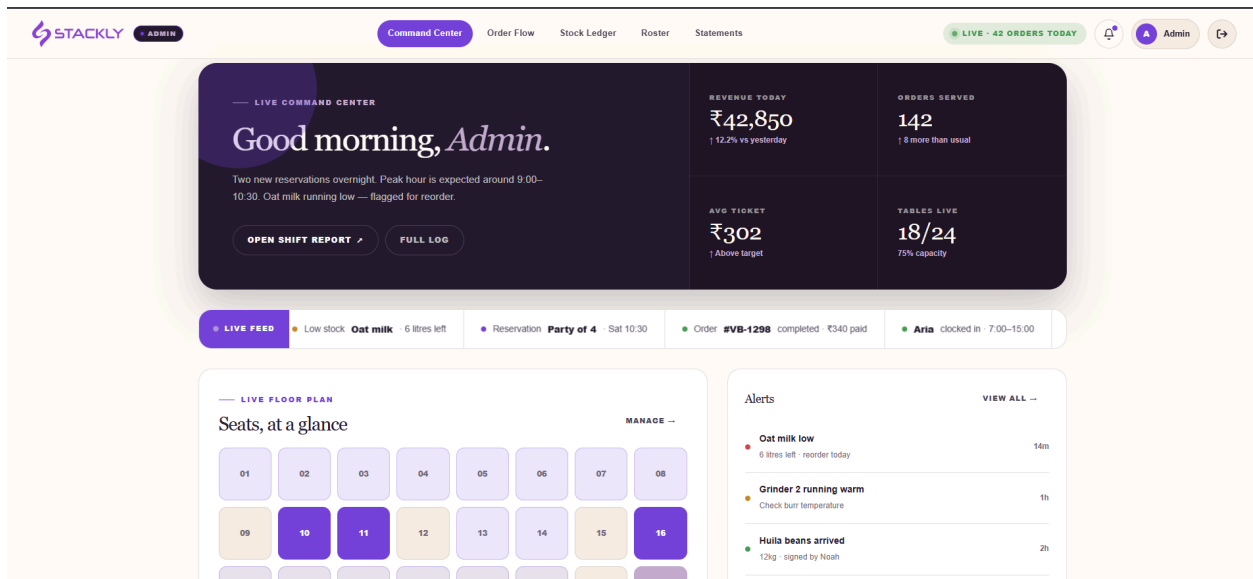
- Financial hero with net profit headline
- Five breakdown rows with progress bars (Staff salaries, Beans & supply, Rent & utilities, Marketing, Other)
- Ledger table with six recent transactions

- Type indicators (Income, Expense, Pending)
- Amount color coding (positive/negative)

Styles: Inline styles within /html/admin.html

Scripts: Embedded JavaScript within /html/admin.html

Assets: /logo.webp



User Dashboard (/html/user.html)

Top Bar Navigation: Displays the brand logo, chip ("Member"), interactive menu links (My Home, My Orders, My Table, Punch Card, My Notes), search input, user chip, logout button, and mobile menu toggle.

Full-Screen Mobile Drawer: Features a full-screen overlay drawer with:

- Brand header and close button
- User greeting
- Five navigation links with numbers and arrows
- User info and logout button
- Body scroll locking

Dynamic Tab Views: Organized into five separate content panels for instant switching:

My Home / Morning Briefing View:

- Editorial briefing with date, greeting, and lede
- Today's Pick sidebar with coffee image and recommendation
- Three facts (Current Streak, Punches, This Month)
- "Cups I've Loved" horizontal reading rail with five favorite drinks
- Two reading cards (From the Bar, A Small Note)

My Orders / Ledger Journal View:

- Single-column reading flow with date dividers
- Five journal entries with time, title, items, amount, and status
- One cancelled entry with strikethrough
- Sticky sidebar with three stats (This Month, Favourite Day, Lifetime Cups)
- Editorial note about ordering patterns

My Table / Planner Spread View:

- Two-week planner strip with 14 day cells
- Today highlighted, booked days marked
- Four table tickets:
 - Next visit (dark card with Window 3)
 - Upcoming visit (Back Room)
 - Two past visits
- Each with seat, date, time, party size, and actions

My Punch Card / Journey Map View:

- Horizontal journey path with 10 stops
- Seven filled, one current, two empty
- Current stop with pulse animation

- Reward stop (free cup) with star
- Goal card with progress bar (70%)
- Activity feed with five recent punches

My Notes / Profile Spread View:

- Profile card (dark) with avatar, name, email, and four stats
- Profile narrative with personal note from the team
- Preferences section with five pref rows:
 - Roast Level (Medium)
 - Milk Choice (Oat milk)
 - Sweetener (None)
 - Caffeine (Extra shot)
 - Visit Time (Mornings)
- Each with progress bar and value

Styles: Inline styles within /html/user.html

Scripts: Embedded JavaScript within /html/user.html

Assets: /logo.webp

Animation & Performance Standards

Hardware-Accelerated Motion: Implements smooth UI transitions and keyframe animations using CSS properties like `transform` and `opacity` to ensure high-framerate, lag-free performance across mobile and desktop devices. The platform uses `transform: translateY(-2px)` for hover effects and `transform: scale(1.02)` for interactive elements.

GSAP & ScrollTrigger Integration: Leverages GSAP and ScrollTrigger for enhanced scroll-driven animations, including staggered entrance animations, floating card effects (continuous yoyo motion), orbit rotations, and scroll-triggered reveals across public-facing pages. These are loaded via CDN and utilized in page-specific JavaScript files.

Non-Blocking UI Rendering: Utilizes lightweight CSS effects and optimized layout structures, ensuring smooth rendering without causing layout shifts or scroll stuttering.

Responsive Micro-Interactions: Elevates the user experience with instant hover and focus feedback—including subtle card scaling, floating animations, orbit rotations, and continuous floating badges—delivering a tactile, app-like feel.

Marquee Performance: Editorial marquees use CSS keyframe animation with `transform: translateX()` for smooth, hardware-accelerated infinite scrolling.

Authentication & Error Handling

Client-Side Security & Validation

Form Validation Rules: Validates input requirements across authentication fields, checking email regex patterns, enforcing minimum 6-character password lengths, verifying role selections, and ensuring password confirmation matching on signup.

Real-time Error Guidance: Prevents default submission on invalid entries, displaying targeted error messages with clear visual feedback. The custom role dropdown (User/Admin) ensures proper role selection before submission.

Local Session Storage: Captures validated credentials upon login, storing the user email into localStorage to retain user identity across pages. The dashboard greeting dynamically displays the stored username.

Role-Based Portal Routing: Evaluates selected account roles (User/Admin) post-validation, dynamically directing users to /html/admin.html or /html/user.html. The role hidden input stores the selected role for processing.

Password Confirmation: On signup, validates that both password fields match before allowing submission, with clear error messaging for mismatched passwords.

Form Submission Prevention: All forms (login, signup, contact, newsletter, reservation) prevent default submission with `event.preventDefault()` and handle validation programmatically, ensuring data integrity before any redirect.

Session Persistence: The `pageshow` event listener ensures form data and session state persist during browser navigation (back/forward), with automatic form reset when returning to authentication pages.

Responsive & Performance Methodologies

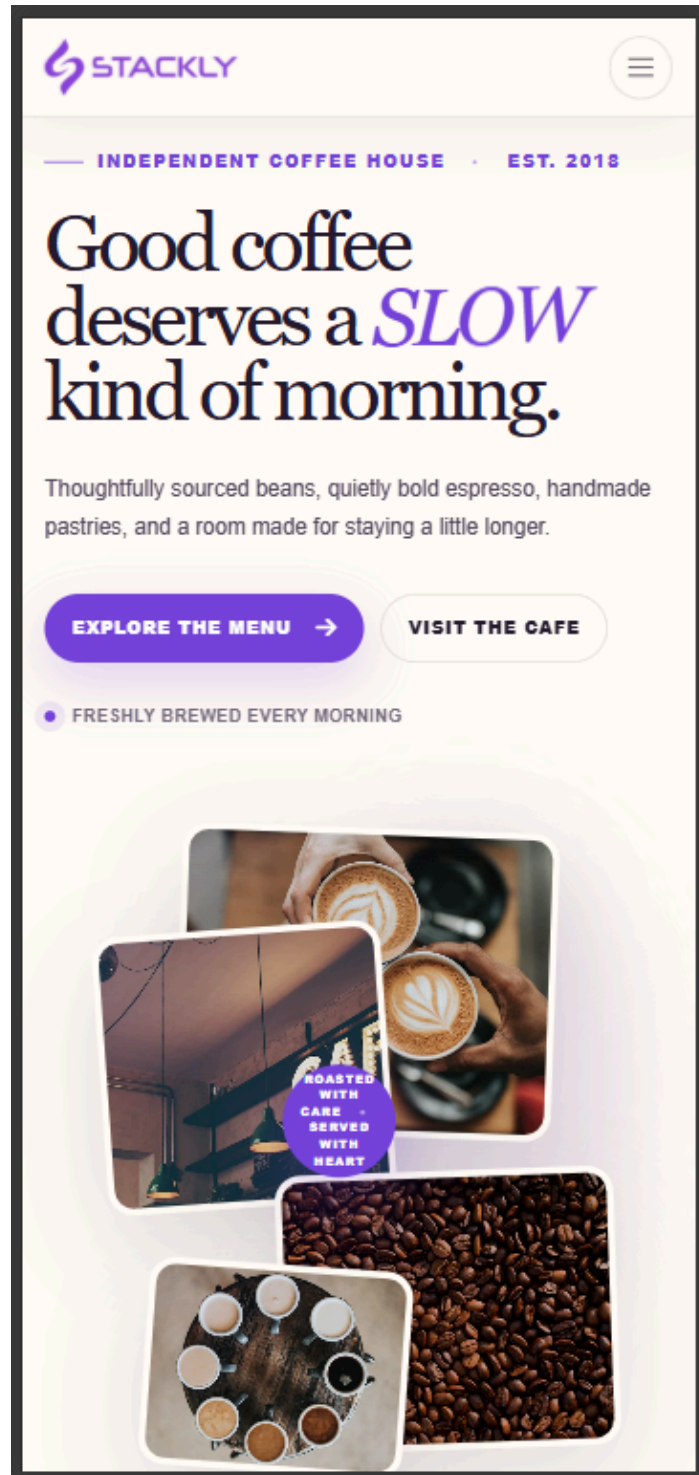
Fluid Layouts & Adaptive Breakpoints: Employs flexible CSS grid structures, flexbox containers, and standard media queries across all viewports (1100px, 900px, 850px, 700px, 480px breakpoints) to ensure seamless UI adaptation from mobile phones up to high-resolution desktop displays. The dashboard sidebar transforms into a slide-out menu on mobile, and complex grids collapse into single-column layouts.

Hardware-Accelerated GSAP Animations: Utilizes lightweight GSAP timelines and ScrollTrigger for high-frame-rate UI entrance transitions, including staggered card reveals and scroll-triggered effects, reducing main-thread execution time and avoiding layout thrashing.

Optimized Asset Delivery: Enhances initial page load performance by utilizing modern WebP image formats across all hero banners, menu illustrations, gallery imagery, and team photos, combined with explicit `loading="lazy"` attributes on heavy graphic elements to defer off-screen image loading until needed.

Dashboard Performance: Implements efficient DOM manipulation and event delegation for view switching, with smooth GSAP transitions between panel sections.

Mobile-First Adaptations: The dashboard sidebar collapses into a slide-out menu on mobile devices, complex grids collapse into single-column layouts, and typography sizes adjust to ensure optimal readability and usability on smaller screens. The hamburger menu provides full-screen mobile navigation with smooth transitions.



SEO & Asset Optimization

Deferred & Asynchronous Resource Loading: Critical external libraries—including GSAP and ScrollTrigger—are pulled via CDN scripts with optimized loading

strategies, while non-critical graphics (brand logos, background banners, decorative images) explicitly declare `loading="lazy"` to eliminate render-blocking delays.

Semantic HTML Markup & Document Metadata: Structured using standard structural tags (`<header>` , `<nav>` , `<aside>` , `<main>` , `<footer>`) with explicit `<title>` tags (e.g., "Stackly Brew — Coffee, Slowly Made", "About — Stackly Brew") and responsive viewport meta elements (`<meta name="viewport" content="width=device-width, initial-scale=1.0">`) on every page to ensure optimal search engine indexing and mobile usability.

Clean URL Structure & Navigation: Implements intuitive, SEO-friendly URL patterns across all pages with consistent internal linking through the global navigation header, ensuring search engines can effectively crawl and index the entire site architecture.

Asset Formats: Leverages optimized WebP media files across all property images, heroes, and background banners, significantly minimizing file weights compared to traditional PNG or JPEG formats while maintaining visual quality.

Future Scalability Insights

While the current frontend architecture delivers a responsive, visually compelling user experience and meets immediate cafe business requirements, enterprise scaling requires full backend integration and enhanced data persistence.

Backend Framework Integration

Transitioning the static HTML, JavaScript, and tab-based architecture into a framework like React, Angular, or Vue.js—paired with a Node.js, Python, or .NET Core API backend—will enable robust state management, modular component reusability, and efficient routing across complex dashboard layouts.

Database & Secure Storage

Migrating temporary localStorage session data (user profiles, role-based permissions, login status, order history) into a relational database such as PostgreSQL or MySQL will securely store customer profiles, order histories, reservation records, loyalty points, and preferences, ensuring data persistence and enabling advanced query capabilities.

RESTful API Architecture

Building a comprehensive RESTful API layer will enable seamless data exchange between the frontend and backend, supporting CRUD operations for orders, reservations, customer profiles, menu items, inventory, and analytics. This will also facilitate future mobile application development and third-party integrations.

Payment & Loyalty Integration

Integrating secure payment processing (Stripe, Square) will enable in-app ordering, digital wallet support, and automated loyalty point tracking. The existing punch card system is designed to support this functionality.

Real-Time Order Tracking

Implementing WebSocket connections will enable real-time order status updates, live queue positions, and instant notifications when orders are ready for pickup.

Multi-Location Support

The architecture can be extended to support multiple cafe locations, enabling customers to select their preferred store, view location-specific menus, and manage reservations across different venues.

Inventory Management

Building automated inventory tracking with supplier integration will enable real-time stock monitoring, automated reordering, and waste reduction analytics.

Customer Analytics Dashboard

Extending the admin dashboard with customer analytics will provide insights into:

- Peak hours and ordering patterns
 - Popular menu items and trends
 - Customer retention and loyalty metrics
 - Revenue forecasting and inventory planning
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